

The Three-Prime Running LCM: Complete Reasoning Record

Erdős Problem #269

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ABSTRACT

The three-prime reciprocal running-LCM kernel has nonsingular minors of every order, uniformly in its third-coordinate layer. A binary floor carry explains its non-separability, exact sampled ranks and sharp uniform approximation obstruction. This is compatible with finite-rank approximation of the original kernel in the summation norm. For the repeated $\{2, 3, 5\}$ series we derive the literal integer tail recurrence, two quadratic bounds, exact denominator clearing and a residue-escape equivalence for dominating caps of size $o(8^a)$. We sharpen the fixed-start residue description and isolate the moving boundary in a value-preserving weighted difference; a literal counterexample shows that absence of floor crossings alone does not force cancellation. No cofinal escape or irrationality theorem is proved for this series. The two-prime comparison reproduces the earlier reduction attributed to Steve Fan, using the classical Hecke–Mahler value theorem. Ordinary proofs, external analytic inputs, supplied formal receipts and finite computations are kept separate.

How the short paper uses this record. The short paper centres on the binary carry and its rank obstruction. Section 4 gives the rank and approximation arguments; Section 3 gives the two-prime comparison. Sections 5–7 recover the literal tails, denominator clearing and window criterion, while Section 8 retains the finite certificates. These are two distinct mathematical tasks: the rank argument excludes a separated representation, and the tail argument identifies the arithmetic condition still needed for irrationality. Neither substitutes for the other.

1 The problem, and what is settled

Let P be a finite set of primes with $|P| \geq 2$, and let $a_1 < a_2 < \dots$ enumerate the positive integers all of whose prime factors lie in P . Erdős Problem #269 asks whether

$$\sum_{n \geq 1} \frac{1}{[a_1, \dots, a_n]}$$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

is irrational, where $[a_1, \dots, a_n]$ is the least common multiple [1, p. 65][2, p. 106]. Bloom's catalogue is the source of the problem number and the status recorded in the cited snapshot [4]. This revision does not make a new claim about the live catalogue status. Theorem 1.1 settles every instance with $|P| = 2$, at the level of transcendence, and Fan posted that argument first [12]; this record leaves the repeated finite cases with $|P| \geq 3$ unresolved.

The restriction $|P| \geq 2$ is necessary. For $P = \{p\}$ the enumeration is $a_n = p^{n-1}$, so $[a_1, \dots, a_n] = p^{n-1}$ and the sum is $p/(p-1)$. For infinite P the sum is always irrational, which Erdős calls a simple exercise [2, p. 106].

Write $\mathcal{R}_P$ for the sum above and $\mathcal{D}_P$ for the de-duplicated sum, in which each distinct value of the running least common multiple contributes its reciprocal once. The two differ because the running value is constant along stretches of the enumeration. In a letter written on 1 January 1973 Erdős recorded that he could prove irrationality of the de-duplicated sum [3, p. 335]. He states that assertion for given primes $p_1, \dots, p_r$ and supplies no proof; for a single prime the de-duplicated sum is again $p/(p-1)$, so the assertion concerns sets of at least two primes. The open question addressed by the catalogue is $\mathcal{R}_P$; its two-prime case is Theorem 1.1.

Theorem 1.1 (two-prime transcendence). *Let p and q be distinct primes. Then $\mathcal{R}_{\{p,q\}}$ and $\mathcal{D}_{\{p,q\}}$ are transcendental.*

Section 3 proves this. Both values are nonconstant polynomials over $\mathbb{Q}$ in one Hecke–Mahler boundary sum A , and transcendence of A is the theorem of Loxton and van der Poorten [11, Theorem 8, p. 40] in the modern form of Bugeaud and Laurent [10, Theorem 1.1]. The evidence class is an ordinary proof in this paper over a cited external theorem, and no Lean declaration formalises it.

Attribution. The supplied forum record credits Steve Fan's post of 26 June 2026 with the running-LCM identity for finite prime sets, the two-channel factorisation, the Hecke–Mahler reduction and its transcendence conclusion [12]. The unrestricted analogue is classical: iterating the prime-exponent maximum rule gives $\text{lcm}(1, \dots, N) = \prod_{t \leq N} t^{\lfloor \log_t N \rfloor}$ over the primes $t \leq N$, equivalently $\log \text{lcm}(1, \dots, N) = \psi(N)$; see Apostol [6] and Montgomery and Vaughan [8]. We reproduce that argument to fix normalisations and distinguish repeated from distinct-height sums, not as a new application of the value theorem. The live thread was inaccessible during this revision; priority is retained from the supplied record rather than re-presented as independently verified correspondence. The scalar transcendence theorem itself is credited separately to its published authors.

What the third prime does. Section 4 isolates the obstruction exactly. After separating row and column factors, the two-prime kernel is an outer product and the three-prime kernel retains one binary floor carry. That carry is enough to produce nonsingular minors of every order, uniformly in the third coordinate, and to make the normalised carry matrix inapproximable by finite separated sums below half its own jump. The minors use only row and column rescaling, the density of the rotations by $\log_r p$ and $\log_r q$ taken separately, and a staircase determinant, and one choice of indices serves every layer of

the literal kernel. Sections 5 to 7 take a second route on the literal $\{2, 3, 5\}$ series: dyadic shells give an integer-coefficient tail recurrence, a strict endpoint clears the smooth part of a hypothetical denominator, and the surviving carry is trapped between a quadratic bound and a residue condition. Section 8 reports the finite evidence and its ceiling, and Section 9 states what remains.

Throughout, p, q, r are pairwise distinct primes. Call n *smooth* when $n = p^i q^j r^k$ for some $i, j, k \geq 0$; this is the smooth lattice value. For $x \geq 1$ write

$$L(x) = \text{lcm}\{n \leq x : n \text{ smooth}\}, \quad H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

the running least common multiple and the pure-power height, where $\lfloor \log_b x \rfloor$ is the largest e with $b^e \leq x$. Since $a_1, \dots, a_n$ are exactly the smooth numbers up to a_n , we have $[a_1, \dots, a_n] = L(a_n)$. The reciprocal of the height at a smooth point is the lattice kernel

$$K(i, j, k) = \frac{1}{H(p^i q^j r^k)}.$$

Here “smooth” always means supported on the fixed prime set, and it is not the varying-bound notion counted by $\Psi(x, y)$ in the Dickman–Hildebrand theory [7]; no smooth-number density asymptotic is used below. The appropriate fixed-support context is Tijdeman–Meijer [23, Sections 1–4]. The modern two-prime treatment of Languasco, Luca, Moree and Togbé [24] also makes the lattice-triangle geometry explicit; its gap estimates are not inputs to our shell bound. For $b \in \{p, q, r\}$ the set of positive powers of b is the b -channel.

The statement of Problem #269 has been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [5]. That is a formal statement of the question up to a rational normalisation, since its Nat-indexed series includes the empty-prefix term, and its rational, irrational and infinite-prime assertions all end in sorry. Kovač and Tao [9] treat several irrationality problems of Erdős for series of unit fractions by elementary means; nothing from that work is used here.

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Notation and proof dependencies

There are two different sums: $\mathcal{R}_p$ counts every supported integer, whereas $\mathcal{D}_p$ counts each height once. From the dyadic section onwards, $S = \mathcal{R}_{\{2,3,5\}}$. Write $P_a = H(2^a)$, $h_a = P_a/2$, s_a for the shell mass and $X_a = h_a \sum_{j \geq a} s_j$. Thus P_a is a boundary height, not a set of primes. The long record’s U_a is the short note’s T_a . The cumulative jump count is $n_a = a + \lfloor a \log_3 2 \rfloor + \lfloor a \log_5 2 \rfloor$; the strict index is $j_a = n_a - 1$ for $a \geq 1$. The long cap is $K(B, a) = \lfloor BQ(n_a) \rfloor$, while the short note uses $K_0(B, a) = 90B(a+1)^2$. Neither is a residue or a hypothetical integer carry. A window has start ℓ , length h , product $W_{\ell, h}$ and forcing $F_{\ell, h}$; zero modulo W is represented by W , not zero.

The rank proof uses only diagonal rescaling and separate one-dimensional density. The arithmetic proof instead uses the actual shell coefficients: summability $\rightarrow$ recurrence $\rightarrow$ denominator clearing $\rightarrow$ the residue criterion. The weighted-shift discussion is a third route, with explicit unproved cancellation hypotheses. No arrow from kernel rank to scalar irrationality is asserted.

2 The finite geometry of the running value

The smooth numbers up to x are indexed by the exponent triples (i, j, k) with $i \leq \lfloor \log_p x \rfloor$, $j \leq \lfloor \log_q x \rfloor$, $k \leq \lfloor \log_r x \rfloor$ and $p^i q^j r^k \leq x$, the smooth prefix index set. Both conditions matter: the coordinate box is strictly larger than the prefix, since a product of three large pure powers can exceed x while each factor does not.

Theorem 2.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then $L(x) = H(x)$.*

Proof. Every smooth $n \leq x$ has exponents bounded by the corresponding integer logarithms, so $n \mid H(x)$ and hence $L(x) \mid H(x)$. Conversely the three pure powers $p^{\lfloor \log_p x \rfloor}$, $q^{\lfloor \log_q x \rfloor}$ and $r^{\lfloor \log_r x \rfloor}$ are themselves smooth numbers not exceeding x , so each divides $L(x)$, and distinct primes have coprime powers, so their product divides $L(x)$ as well. $\square$

At $(p, q, r) = (2, 3, 5)$ the first ten values are

x	1	2	3	4	5	6	7	8	9	10
$L(x)$	1	2	6	12	60	60	60	120	360	360

So $L(6) = 4 \cdot 3 \cdot 5 = 60$, which exceeds 6: the running value at a smooth cutoff already contains powers of the other two primes that the cutoff itself does not. That small fact is the mechanism behind Section 4. Also $H(x) \leq x^3$, since each of the three factors is a power of its base not exceeding x ; this is the cubic majorant, and the exponent is the number of generating primes.

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x \rfloor = \lfloor \log_b y \rfloor$ for each of $b = p, q, r$, the cell relation. By Theorem 2.1 the running value depends on x only through the three integer logarithms, so it is constant on cells and moves only where one logarithm moves.

Proposition 2.2 (constancy and jump ratios). *If $x, y \geq 1$ lie in the same logarithmic cell then $L(x) = L(y)$, and the same holds for the kernel at two smooth points of one cell. If $\lfloor \log_p y \rfloor = \lfloor \log_p x \rfloor + 1$ while the other two logarithms agree, then $L(y) = p L(x)$, and similarly with q or r in place of p .*

Proof. Both parts are immediate from Theorem 2.1: the height depends on x only through the three integer logarithms, and advancing one of them multiplies exactly one factor by its base. $\square$

Proposition 2.3 (jump count). *Let $n \geq 0$. The set of the first n positive powers of p , of q and of r has exactly $3n$ elements, and adjoining the common origin 1 gives exactly $3n + 1$.*

Proof. Within one channel the powers $b, b^2, \dots, b^n$ are distinct because $b \geq 2$. Across two channels a common value would be a positive power of two distinct primes, which unique factorisation forbids. Finally 1 is not a positive power of any prime. $\square$

Two finite statements are recorded here because they are the exact shape of the counting used later. Write $\mathcal{B}(h_p, h_q, h_r)$ for the box of exponent triples with $i \leq h_p, j \leq h_q, k \leq h_r$, the exponent box, and $F(H)$ for the set of its points whose height equals H , the height fibre.

Proposition 2.4 (finite normal form). *For every box $\mathcal{B}$, $\sum_{(i,j,k) \in \mathcal{B}} K(i,j,k) = \sum_H \#F(H)/H$, the outer sum ranging over the heights attained on $\mathcal{B}$.*

Proof. Partition $\mathcal{B}$ into the fibres of the height map. On $F(H)$ every summand is $1/H$, so the fibre contributes $\#F(H)/H$. $\square$

Now fix an interval $[\lambda, \eta)$ and write $\mathcal{S}$ for the exponent triples of $\mathcal{B}(h_p, h_q, h_r)$ whose smooth value lies in it, the smooth exponent shell.

Lemma 2.5 (uniqueness in a short interval). *Let $b \geq 1$ and $\eta \leq b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda, \eta)$ then $a = a'$.*

Proof. If $a < a'$ then $\eta \leq b\lambda \leq b^{a+1}w \leq b^{a'}w < \eta$, which is impossible; the case $a > a'$ is symmetric. $\square$

Proposition 2.6 (projection and the quadratic shell bound). *If $\eta \leq r\lambda$ then $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$, and if $\eta \leq p\lambda$ then $\#\mathcal{S} \leq (h_q + 1)(h_r + 1)$. If moreover $\eta \leq r\lambda$ and $h_p \leq h_q \leq h_r$ with $h_p + h_q + h_r = j$, then $9\#\mathcal{S} \leq (j + 3)^2$.*

Proof. Suppose $\eta \leq r\lambda$. If two triples of $\mathcal{S}$ agree in their first two coordinates, Lemma 2.5 with $b = r$ and $w = p^i q^j$ forces their third coordinates to agree, so the projection forgetting the third coordinate is injective on $\mathcal{S}$ and its image lies in a rectangle with $(h_p + 1)(h_q + 1)$ points. The other case is the same with the first coordinate projected away. Under the sorting hypothesis the two surviving coordinates are the two smallest, so it suffices that $a \leq b \leq c$ with $a + b + c = j$ gives $9(a + 1)(b + 1) \leq (j + 3)^2$. From $a \leq b \leq c$ we get $a + 2b \leq j$, so it is enough that $9(a + 1)(b + 1) \leq (a + 2b + 3)^2$; writing $b = a + d$ with $d \geq 0$, the difference of the two sides is $d(3a + 4d + 3) \geq 0$. $\square$

3 One Hecke–Mahler value controls both two-prime sums

Temporarily let $P = \{p, q\}$ with $p < q$, and write $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$, which is the running least common multiple of the $\{p, q\}$ -smooth numbers up to t by the argument of Theorem 2.1 with one coordinate omitted. The de-duplicated sum retains the initial value 1 and one reciprocal for every later distinct running value, so

$$\mathcal{D}_{\{p,q\}} = 1 + \sum_{t \in \{p, p^2, \dots\} \cup \{q, q^2, \dots\}} \frac{1}{L_{p,q}(t)}, \quad \mathcal{R}_{\{p,q\}} = \sum_{i,j \geq 0} \frac{1}{L_{p,q}(p^i q^j)}.$$

Proof of Theorem 1.1. Set

$$\theta = \frac{\log p}{\log q}, \quad x = \frac{1}{p}, \quad y = \frac{1}{q}, \quad m_n = \lfloor n\theta \rfloor, \quad \delta_n = m_{n+1} - m_n.$$

Here $0 < \theta < 1$, and θ is irrational, since a rational value would give $p^b = q^a$ for positive integers a, b . Consequently $\delta_n \in \{0, 1\}$. Put

$$A = \sum_{n \geq 0} x^n y^{m_n}, \quad B_* = \sum_{n \geq 0} \delta_n x^n y^{m_{n+1}}.$$

The initial value and the p -channel contribute A , since $L_{p,q}(p^n) = p^n q^{m_n}$. A power of q lies strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, it is then $q^{m_{n+1}}$, and its post-jump reciprocal is $x^n y^{m_{n+1}}$; so the q -channel contributes B_* and $\mathcal{D}_{\{p,q\}} = A + B_*$. All these series converge absolutely.

Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$, an index shift gives $A - 1 - xA = x(y - 1)B_*/y$, so $B_* = ((p - 1)A - p)/(1 - q)$ and

$$\mathcal{D}_{\{p,q\}} = \frac{(q - p)A + p}{q - 1}. \quad (1)$$

At a smooth point, $\log_p(p^i q^j) = i + j/\theta$ and $\log_q(p^i q^j) = j + i\theta$, so $L_{p,q}(p^i q^j) = p^{i + \lfloor j/\theta \rfloor} q^{j + m_i}$ and absolute convergence permits the factorisation

$$\mathcal{R}_{\{p,q\}} = AC, \quad C = \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor}.$$

For $j \geq 1$ put $n = \lfloor j/\theta \rfloor$. Then $n > j/\theta - 1$ gives $n\theta > j - \theta > j - 1$, and $n + 1 > j/\theta$ gives $(n + 1)\theta > j$, while $n\theta \leq j$ and $(n + 1)\theta < j + \theta < j + 1$; hence $m_n = j - 1$, $m_{n+1} = j$ and $\delta_n = 1$. Conversely every n with $\delta_n = 1$ arises from the unique $j = m_{n+1}$. Therefore $C = 1 + B_*$ and

$$\mathcal{R}_{\{p,q\}} = \frac{(p + q - 1)A - (p - 1)A^2}{q - 1}. \quad (2)$$

It remains to prove that A is transcendental. For the Hecke–Mahler series

$$F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{\lfloor n\theta \rfloor} x^n y^k$$

a finite geometric sum gives $((1 - y)/y)F_\theta(x, y) = x/(1 - x) - (A - 1)$, that is

$$A = \frac{1}{1 - x} - \frac{1 - y}{y} F_\theta(x, y). \quad (3)$$

Bugeaud and Laurent's Theorem 1.1 states, in particular, that $F_\theta(\beta, \alpha)$ is transcendental when $\theta \in (0, 1)$ is irrational, α and β are nonzero algebraic numbers, $|\beta| < 1$ and $|\beta\alpha^\theta| < 1$ [10, Theorem 1.1]; the $\rho = 0$ case used here goes back to Loxton and van der Poorten [11, Theorem 8, p. 40]. Take $(\beta, \alpha) = (x, y)$: then $|\beta| = 1/p < 1$ and

$$|xy^\theta| = \frac{1}{p} \left(\frac{1}{q} \right)^{\log p / \log q} = \frac{1}{p^2} < 1.$$

So $F_\theta(x, y)$ is transcendental, and (3) makes A transcendental. The coefficient of A in (1) is $(q - p)/(q - 1) \neq 0$ and the coefficient of A^2 in (2) is $-(p - 1)/(q - 1) \neq 0$, both rational. If either value were algebraic, its identity would exhibit A as a root of a nonzero polynomial over the algebraic numbers. $\square$

The two identities are separately labelled because they carry different weight. A product of two transcendental numbers need not be transcendental, so the factorisation $\mathcal{R} = AC$ alone proves nothing; the quadratic identity (2) in the single value A is what settles the repeated sum. A third prime replaces the single Beatty boundary by a genuinely two-dimensional ordering problem, and the next section makes that failure exact.

4 The rank phase transition

One might hope to write the kernel as $f(i)g(j)h(k)$ and reduce the problem to one-dimensional criteria. At two generators that hope is exactly correct, and at three it fails at every finite order.

Proposition 4.1 (two generators separate). *For real $p, q > 1$, with $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$ as above, and all integers $i, j \geq 0$, the two-prime kernel $K_2(i, j) = 1/L_{p,q}(p^i q^j)$ is the outer product*

$$K_2(i, j) = (p^i q^{\lfloor \log_q p^i \rfloor})^{-1} (p^{\lfloor \log_p q^j \rfloor} q^j)^{-1},$$

so every two-by-two minor of K_2 vanishes.

Proof. Since i and j are integers, $\lfloor \log_p(p^i q^j) \rfloor = i + \lfloor \log_p q^j \rfloor$ and $\lfloor \log_q(p^i q^j) \rfloor = j + \lfloor \log_q p^i \rfloor$. Hence $L_{p,q}(p^i q^j)$ is the product of $p^i q^{\lfloor \log_q p^i \rfloor}$, which depends on i alone, and $p^{\lfloor \log_p q^j \rfloor} q^j$, which depends on j alone. A matrix whose entries are a product of a row function and a column function has vanishing two-by-two minors. For distinct primes p, q the product $L_{p,q}$ is the running least common multiple by the argument of Theorem 2.1. $\square$

At three generators the smallest rectangle already fails to factor. A factorisation $f(i)g(j)h(k)$ would force $K(0, 0, 0)K(1, 1, 0) = K(1, 0, 0)K(0, 1, 0)$.

Proposition 4.2 (non-separability at $\{2, 3, 5\}$). *With $(p, q, r) = (2, 3, 5)$,*

$$\det \begin{pmatrix} K(0, 0, 0) & K(0, 1, 0) \\ K(1, 0, 0) & K(1, 1, 0) \end{pmatrix} = \det \begin{pmatrix} 1 & 1/6 \\ 1/2 & 1/60 \end{pmatrix} = -\frac{1}{15} \neq 0.$$

Proof. The four values are computed from $H(1) = 1$, $H(2) = 2$, $H(3) = 2 \cdot 3 = 6$ and $H(6) = 4 \cdot 3 \cdot 5 = 60$, so the determinant is $1/60 - 1/12 = -1/15$. $\square$

Theorem 4.3 (arbitrary-order non-separability). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 1$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell : \mathbb{N} \rightarrow \mathbb{Q}$ and $G_\ell : \mathbb{N}^2 \rightarrow \mathbb{Q}$, $0 \leq \ell < d$, satisfying $K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k)$ for all i, j, k .

Proof. Put $\alpha = \log_r p$, $\beta = \log_r q$, $x_i = \{i\alpha\}$ and $y_j = \{j\beta\}$. The three height exponents are

$$v_p(H(p^i q^j r^k)) = i + \lfloor j \log_p q + k \log_p r \rfloor,$$

$$v_q(H(p^i q^j r^k)) = j + \lfloor i \log_q p + k \log_q r \rfloor,$$

$$v_r(H(p^i q^j r^k)) = k + \lfloor i\alpha \rfloor + \lfloor j\beta \rfloor + \lfloor x_i + y_j \rfloor.$$

Every term except the last floor depends on i and k alone or on j and k alone, so with positive rational $R_i(k)$ and $C_j(k)$,

$$K(i, j, k) = R_i(k) C_j(k) t^{\lfloor x_i + y_j \rfloor}, \quad t = r^{-1}. \quad (4)$$

The remaining matrix is independent of k , and $x_i + y_j \in [0, 2)$, so its entries are 1 and t .

Each of α and β is irrational, since a rational value would give an equality of positive powers of distinct primes, so each fractional-part orbit is dense in $(0, 1)$ and each is injective. No simultaneous density of the pair of rotations is required. First choose indices with $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. For the columns write $s_b = 1 - y_{J(b)}$, and use density of the second orbit to choose

$$s_0 \in (0, x_{I(0)}), \quad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals are disjoint, so the $J(b)$ are distinct. Their strict endpoints ensure that no selected entry lies on a threshold, and $x_{I(a)} + y_{J(b)} \geq 1$ holds exactly when $b \leq a$. Dividing row a by $R_{I(a)}(k)$ and column b by $C_{J(b)}(k)$ therefore leaves

$$C_n(t) = \begin{pmatrix} t & 1 & 1 & \dots & 1 \\ t & t & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ t & t & \dots & t & 1 \\ t & t & \dots & t & t \end{pmatrix}, \quad \det C_n(t) = t(t-1)^{n-1} \neq 0,$$

the determinant following by subtracting from each row its predecessor, working upwards from the last. Before the division, the determinant equals

$$\det C_n(t) \prod_{a < n} R_{I(a)}(k) \prod_{b < n} C_{J(b)}(k),$$

which is nonzero for every k ; this also explains why the same indices work in every layer.

For the last assertion, suppose that a representation with d summands exists and fix k . On the selected rows $I(0), \dots, I(d)$ and columns $J(0), \dots, J(d)$ the resulting matrix factors as

$$(f_\ell(I(a)))_{0 \leq a \leq d, \ell < d} (G_\ell(J(b), k))_{\ell < d, 0 \leq b \leq d}.$$

It has rank at most d and hence zero determinant, contradicting the nonzero minor of order $d+1$. $\square$

Index selection is essential. The leading minors of the $\{2, 3, 5\}$ kernel are not a witness: row three is $1/120$ times row zero for $j \leq 3$ (checked), and the proportionality fails at $j = 4$ (checked). A proof that read off leading minors alone would be false.

The same threshold description determines every finite sampled rank, rather than only producing one nonsingular minor.

Proposition 4.4 (finite sampled cut rank). *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq h \leq m$, let v_h be the length- m column whose first h entries are 1 and whose remaining entries are c . If the distinct columns of a matrix are the v_h with h in a nonempty set $E \subseteq \{0, \dots, m\}$, then its rank is*

$$|E| - \mathbf{1}_{\{0, m\} \subseteq E}.$$

Proof. Write $E = \{h_1 < \dots < h_s\}$. The $s - 1$ consecutive differences are

$$v_{h_{a+1}} - v_{h_a} = (1 - c)\mathbf{1}_{\{h_a, \dots, h_{a+1}-1\}},$$

so they are linearly independent because their nonempty supports are disjoint. If $h_1 > 0$ or $h_s < m$, those supports miss a coordinate on which v_{h_1} is nonzero, and adjoining v_{h_1} gives rank s . If $h_1 = 0$ and $h_s = m$, the differences sum to $(1 - c)\mathbf{1}$ while $v_0 = c\mathbf{1}$, so v_0 is already in their span and the rank is $s - 1$. $\square$

To apply the proposition, sort any chosen row phases x_i . Each normalised column is a threshold column v_h , where h is the number of sampled phases strictly below $1 - y_j$. Repeated threshold positions give proportional columns before column normalisation, and the nonzero row and column factors in (4) do not change rank. Thus the displayed formula is the exact rank of every finite sample.

Attribution and reach. Fan recorded the qualitative obstruction for $|P| \geq 3$ in his comment of 26 June 2026 [12], observing that the two-prime argument does not seem to generalise immediately. The quantitative form above, its uniformity in the third coordinate, and its formalisation are proved here. Theorem 4.3 excludes exact separations of the displayed finite-sum form; it is not an independence, irrationality or transcendence statement, and it settles no instance of Erdős #269.

4.1 A sharp uniform obstruction for the normalised carry matrix

A fixed nonzero determinant remains nonzero under sufficiently small perturbations. What the preceding argument does not provide is one error threshold that works uniformly for the entire infinite matrix and all orders. The next theorem replaces exactness by a uniform distance, and the constant it produces is sharp. It concerns the normalised carry matrix

$$C(i, j) = t^{\lfloor x_i + y_j \rfloor}, \quad x_i = \{i \log_r p\}, \quad y_j = \{j \log_r q\}, \quad t = r^{-1}, \quad (5)$$

which is the factor left in (4) after the row and column factors are divided out. Say that a real matrix A on $\mathbb{N} \times \mathbb{N}$ has *finite separated rank* when all of its columns lie in one finite-dimensional space of real sequences, equivalently when $A(i, j) = \sum_{\ell < d} f_\ell(i) g_\ell(j)$ for some finite d and some sequences f_ℓ, g_ℓ , with no continuity or boundedness assumed.

Theorem 4.5 (sharp uniform separated approximation). *Let p, q, r be pairwise distinct primes and let C be as in (5). Then*

$$\inf_A \sup_{i, j \geq 0} |C(i, j) - A(i, j)| = \frac{1 - t}{2} = \frac{r - 1}{2r},$$

the infimum being over all matrices A of finite separated rank, and it is attained by the constant matrix of value $(1 + t)/2$.

Proof. Every entry of C lies in $\{1, t\}$, and $C(i, j) = t$ exactly when $x_i + y_j \geq 1$. Fix $j \neq k$. The numbers y_j are pairwise distinct, since $\log_r q$ is irrational, so we may assume $y_j < y_k$, and then $0 \leq 1 - y_k < 1 - y_j \leq 1$. Density of the orbit (x_i) in $(0, 1)$ supplies an index

i with $1 - y_k < x_i < 1 - y_j$, and at that row the two columns carry the entries t and 1. Hence any two distinct columns of C are at sup-distance exactly $1 - t$.

Let A have finite separated rank and put $\varepsilon = \sup_{i,j} |C(i,j) - A(i,j)|$. Suppose $\varepsilon < (1 - t)/2$. Each column A_j then satisfies $\|A_j\|_\infty \leq 1 + \varepsilon$, so all columns of A lie in $W = V \cap \ell^\infty$, where V is the finite-dimensional space containing the columns of A ; the space W is a subspace of V , hence finite-dimensional, and $\|\cdot\|_\infty$ is a genuine norm on it. By the triangle inequality, distinct columns satisfy

$$\|A_j - A_k\|_\infty \geq \|C_j - C_k\|_\infty - 2\varepsilon = 1 - t - 2\varepsilon > 0.$$

The compactness input is the standard fact that closed bounded subsets of a finite-dimensional real normed space are compact. The use of $W = V \cap \ell^\infty$ is essential: the individual separated row factors need not be bounded. The columns $A_j, j \geq 0$, are therefore infinitely many points of W , all of norm at most $1 + \varepsilon$ and pairwise separated by a fixed positive distance. A bounded subset of a finite-dimensional normed space is totally bounded, so it contains no infinite uniformly separated family. This contradiction gives $\varepsilon \geq (1 - t)/2$ for every A of finite separated rank.

For sharpness take $A(i,j) = (1 + t)/2$, which has separated rank one; every entry of C is at distance exactly $(1 - t)/2$ from it. $\square$

The compactness step is standard, and the statement is recorded here for this kernel without a claim of technique. Three qualifications fix its reach. It is a statement about the normalised carry matrix (5): the original kernel carries positive row and column factors that decay, so the conclusion transfers to a weighted uniform norm relative to those factors and not to the unweighted sup norm of K . It bounds approximation of the whole infinite matrix, and on any finite range a separated approximant of small rank exists. And it says nothing about approximating the scalar sum, so it yields no irrationality conclusion. What it does add to Theorem 4.3 is robustness: no finite separated model of the carry, however chosen, gets uniformly closer than half a carry jump. The threshold is exact in both directions, since rank one attains it.

The original kernel behaves differently in the summation norm because its row and column factors decay. Let $K^{(N)}(i,j,k) = K(i,j,k)$ for $i < N$ and $K^{(N)}(i,j,k) = 0$ otherwise. This is a sum of at most N terms separated between i and (j,k) . Since $H(x) > x^3/(pqr)$ for $x > 0$, geometric summation gives

$$\sum_{i,j,k \geq 0} |K(i,j,k) - K^{(N)}(i,j,k)| \leq \frac{pqr p^{-3N}}{(1 - p^{-3})(1 - q^{-3})(1 - r^{-3})}. \quad (6)$$

Hence the original summable kernel has finite separated-rank approximants in ℓ^1 , even though its normalised carry matrix has the sharp uniform barrier of Theorem 4.5. This norm distinction is why neither statement decides the arithmetic nature of the scalar sum.

5 Dyadic blocks and the literal infinite tail

For the rest of the note set $P = \{2, 3, 5\}$ and $S = \mathcal{R}_P$, and write $H_a = H(2^a)$ and $h_a = H_a/2$, so that $h_0 = 1/2$ while h_a is a positive integer for $a \geq 1$. Define the dyadic shell

mass, its tail and its normalisation by

$$s_a = \sum_{\substack{i,j,k \geq 0 \\ 2^a \leq 2^i 3^j 5^k < 2^{a+1}}} \frac{1}{H(2^i 3^j 5^k)}, \quad U_a = \sum_{j \geq a} s_j, \quad X_a = h_a U_a. \quad (7)$$

Compress the jump word of Section 2 into the blocks cut out by consecutive powers of two: block a starts just after 2^a , contains every pure 3- or 5-power strictly between 2^a and 2^{a+1} , and ends with the jump at 2^{a+1} . A channel cannot occur twice inside one block, because two powers of the same base $b \geq 2$ inside an interval of ratio $2 \leq b$ must coincide; this is the checked internal-power uniqueness lemma for the internal-power predicate. Let I_a list the internal jumps (p, e) with $p \in \{3, 5\}$ and $2^a < p^e < 2^{a+1}$, in increasing order.

Proposition 5.1 (the dyadic block alphabet). *For every a ,*

$$b_a = \frac{H_{a+1}}{H_a} = 2 \prod_{(p,e) \in I_a} p \in \{2, 6, 10, 30\}, \quad \text{so} \quad 2 \leq b_a \leq 30. \quad (8)$$

Proof. Internal-power uniqueness leaves two independent yes-or-no choices, one for the 3-channel and one for the 5-channel. Multiplying the terminal factor 2 by the selected channel factors gives exactly the four displayed cases. $\square$

For $p \in P$ with q, r the other two primes, put

$$A_p(e) = \#\{(i, j) \in \mathbb{N}^2 : q^i r^j < p^e\}, \quad C_p(e) = \sum_{u=1}^e A_p(u), \quad C_p(0) = 0,$$

and for $(p, e) \in I_a$ let $\sigma_a(p, e)$ be the product of the channels of the later internal jumps of block a . The *ordered block digit* is

$$m_a = A_2(a+1) + \sum_{(p,e) \in I_a} (p-1) \sigma_a(p, e) (C_p(e) - C_2(a)) \quad (a \geq 1), \quad m_0 = 1, \quad (9)$$

the integer implemented by the pinned checker and the checked ordered digit. The first four pairs (b_a, m_a) from $a = 1$ are $(6, 4)$, $(10, 7)$, $(6, 7)$, $(30, 65)$.

Theorem 5.2 (the literal shell recurrence). *The shell masses are summable and $S = \sum_{a \geq 0} s_a$. For every $a \geq 0$,*

$$m_a = h_{a+1} s_a \in \mathbb{N}_{>0}, \quad X_{a+1} = b_a X_a - m_a, \quad X_a = \sum_{j \geq a} \frac{m_j}{b_a b_{a+1} \cdots b_j}. \quad (10)$$

Proof. Every exponent of 3 or 5 in the a th shell is at most a , and for each such pair Lemma 2.5 leaves at most one exponent of 2 placing the point in $[2^a, 2^{a+1})$. Each height is at least 2^a and the shell contains 2^a , so $0 < s_a \leq (a+1)^2 2^{-a}$. The majorant is summable, which justifies every tail splitting below, and unique prime factorisation identifies $\sum_a s_a$ with the original repeated series.

Write the internal jumps as $t_\ell = p_\ell^{e_\ell}$, $1 \leq \ell \leq v$, and set $t_0 = 2^a$, $t_{v+1} = 2^{a+1}$. Let $N(t)$ count smooth positive integers strictly below t and put $n_\ell = N(t_\ell)$. On $[t_\ell, t_{\ell+1})$ the height is $H_a \prod_{u \leq \ell} p_u$, and the terminal jump has factor two, so

$$h_{a+1}s_a = \sum_{\ell=0}^v (n_{\ell+1} - n_\ell) \prod_{u > \ell} p_u = n_{v+1} - n_0 + \sum_{\ell=1}^v (p_\ell - 1) \left(\prod_{u > \ell} p_u \right) (n_\ell - n_0),$$

the second equality being finite summation by parts. Counting by the exponent of p gives $N(p^e) = C_p(e)$, so the right-hand side is exactly (9); at $a = 0$ the shell is the single point 1, giving $m_0 = 1$. Positivity follows from $s_a > 0$. Splitting $U_a = s_a + U_{a+1}$ and multiplying by h_a gives the recurrence, and $b_a \cdots b_j = H_{j+1} / H_a$ with $m_j = h_{j+1}s_j$ gives $m_j / (b_a \cdots b_j) = h_a s_j$, whose summation is the last identity. $\square$

Lemma 5.3 (the literal weighted lattice triangle). *Put $\lambda_3 = \log_2 3$, $\lambda_5 = \log_2 5$ and $\theta_p = 1/\lambda_p$ for $p = 3, 5$. For $j, k \geq 0$ write $w_{j,k} = j\lambda_3 + k\lambda_5$ and $t_{j,k} = \{w_{j,k}\}$. With $P_a = H(2^a)$, the actual shell numerator is*

$$m_a = \sum_{\substack{j,k \geq 0 \\ w_{j,k} < a+1}} 3^{\lfloor (a+1)\theta_3 \rfloor - \lfloor (a+t_{j,k})\theta_3 \rfloor} 5^{\lfloor (a+1)\theta_5 \rfloor - \lfloor (a+t_{j,k})\theta_5 \rfloor}.$$

Every summand is in $\{1, 3, 5, 15\}$. In particular, for $a \geq 0$,

$$(\lfloor a/(2\lambda_3) \rfloor + 1)(\lfloor a/(2\lambda_5) \rfloor + 1) \leq m_a \leq 15(a+1)^2.$$

Thus $m_a = \Theta((a+1)^2)$ and the numerator sequence is unbounded. These are integer forcing numerators, not positional digits restricted to $\{0, \dots, b_a - 1\}$.

Proof. For each pair with $w_{j,k} < a+1$, exactly one exponent $i = a - \lfloor w_{j,k} \rfloor \geq 0$ puts $2^i 3^j 5^k = 2^{a+t_{j,k}}$ in $[2^a, 2^{a+1})$. This is a bijection onto the literal shell. Substituting in $P_{a+1} / (2H(2^i 3^j 5^k))$ cancels the two-power and gives the displayed weight. Each remaining floor increment is zero or one since $0 < \theta_p < 1$. Finally $\lambda_3, \lambda_5 > 1$ implies $j, k \leq a$ for every summation pair, proving the upper bound. For the lower bound, restrict to $0 \leq j \leq \lfloor a/(2\lambda_3) \rfloor$ and $0 \leq k \leq \lfloor a/(2\lambda_5) \rfloor$. Then $w_{j,k} \leq a$ and each weight is at least one. $\square$

This is an exact reindexing, not a polynomial formula for m_a . It isolates the moving triangular boundary as well as the two floor weights. The displayed proof and the finite checks accompanying this revision are not claimed as a new Lean declaration.

Since $h_0 = 1/2$, the last identity at $a = 0$ writes $S/2 = X_0$ as the Cantor series $\sum_{j \geq 0} m_j / (b_0 \cdots b_j)$, and X_a is its normalised tail. Such tails, with their integer carry recurrence, are the classical tool for the rationality of Cantor series. Erdős and Straus characterise rationality by integer carries [13, Theorem 2.1 and (2.4), pp. 85–86], and Hančl and Tijdeman give a practical form of that criterion through tails R_n with $a_n R_n = b_n + R_{n+1}$, which is the recurrence above in their notation [14, §§2–3 and Theorem 3.1, pp. 372–375]. Koutsoukou-Argyaki and Li formalised the Erdős–Straus criteria in Isabelle/HOL [15]. Both criteria assume that the n th numerator is $o(a_{n-1}a_n)$, where a_n is the n th base. Here $m_a \geq 1$ and $b_{a-1}b_a \leq 900$, so neither criterion applies. The passage from rationality to

integral tails is the standard denominator clearing, proved for this series in Section 6; the work specific to this series is the literal forcing with its multiplicities, the strict endpoint normalisation and the explicit bounds. For polynomial Cantor data, Hančl–Tijdeman [20, Theorems 2.2 and 3.1] give a polynomial cancellation criterion and a division mechanism; their nonconstant polynomial radix is not our bounded (b_a) . Their separate Theorem 4.2 permits general integer radices, but assumes an exact finite-product decomposition with an integer transformed numerator $o(b_a)$. Here boundedness would force that transformed numerator to vanish eventually. Such a decomposition has not been proved for the literal m_a ; the bound $m_a = O(a^2)$ does not produce one, nor does it force an integer carry to be a polynomial in the floor coordinates.

Proposition 5.4 (integral state or cofinal separation). *For every integer $B \geq 1$, either $BX_a \in \mathbb{Z}$ for some $a \geq 0$ and every later a , or for every a_0 there is $a \geq a_0$ with $|BX_a - z| \geq 1/31$ for every $z \in \mathbb{Z}$.*

Proof. Set $Y_a = BX_a$, so $Y_{a+1} = b_a Y_a - Bm_a$. If cofinal separation fails, there are A and integers z_a with $e_a = Y_a - z_a$ and $|e_a| < 1/31$ for all $a \geq A$. The recurrence gives $z_{a+1} - b_a z_a + Bm_a = b_a e_a - e_{a+1}$. The left side is an integer and the right side has absolute value below $(30 + 1)/31 = 1$, so both vanish and $e_{a+1} = b_a e_a$. Hence $|e_{A+k}| \geq 2^k |e_A|$ for every k while $|e_{A+k}| < 1/31$, which forces $e_A = 0$. Once Y_A is integral, the integer recurrence makes every later Y_a integral. $\square$

The supplied formal source records the $B = 1$ instance. Multiplying the forcing by the integer B gives the ordinary proof of the scaled statement above, without requiring a new infinite-tail identity. Neither branch is excluded. For comparison, Erdős–Taylor [16, Theorem 1, p. 600] prove a countability statement for increasing integer sequences with bounded successive ratios; Fan [17, Lemma 3.1] gives the unbounded-sequence version. For $g_a = Bh_a$ our integer-ratio argument is stronger: convergence of $\text{dist}(g_a \zeta, \mathbb{Z})$ to zero forces eventual integrality. Applying that statement to the particular $\zeta = S$ still requires excluding the integral branch.

6 A quadratic tail bound and denominator cancellation

Put

$$n_a = a + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor, \quad Q(n) = \frac{n^2 + 8n + 18}{9},$$

so n_a is the sum of the three height exponents at 2^a .

Theorem 6.1 (quadratic bound for the actual tail). *For every $a \geq 0$, $0 < X_a \leq Q(n_a)$.*

Proof. Partition the smooth integers $x \geq 2^a$ by their height vector

$$(A, B, C) = (\lfloor \log_2 x \rfloor, \lfloor \log_3 x \rfloor, \lfloor \log_5 x \rfloor).$$

The cell of a vector is the interval $[\lambda, \eta)$ with $\lambda = \max(2^A, 3^B, 5^C)$ and $\eta = \min(2^{A+1}, 3^{B+1}, 5^{C+1})$, so $\eta \leq 2^{A+1} \leq 2\lambda$ and, by Lemma 2.5, fixing the exponents of 3 and 5 leaves at most one exponent of 2. Since $A \geq B \geq C$, the cell contains at most

$$(B+1)(C+1) \leq \frac{(A+B+C+3)^2}{9}$$

smooth points: writing $u = B + 1 \geq v = C + 1$ and using $A + 1 \geq u$, the difference $(2u + v)^2 - 9uv = (u - v)(4u - v)$ is nonnegative. Unique factorisation identifies these exponent triples with distinct smooth integers.

As the cutoff increases all three height exponents are nondecreasing, so two nonempty cells with the same exponent sum are the same cell, and there is at most one nonempty cell of each rank. Every cell above 2^a has rank at least n_a , and a cell of rank $n_a + k$ has height at least $H_a 2^k$, since each of the k extra prime factors is at least 2. Nonnegative summation over ranks, allowing empty ones, gives

$$X_a \leq \frac{1}{18} \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{2^k} = \frac{n_a^2 + 8n_a + 18}{9},$$

the evaluation using the geometric moments $\sum 2^{-k} = 2$, $\sum k 2^{-k} = 2$ and $\sum k^2 2^{-k} = 6$. Positivity follows from the shell at 2^a . $\square$

The rank here is the sum of the three fixed height exponents, not a varying-smoothness density parameter. The estimate uses only unique factorisation, the short-cell counting bound and geometric moments; it does not invoke a Dickman–Hildebrand asymptotic or a Hecke–Mahler theorem.

This is the analytic content of the section: each additional height rank costs a geometric factor while its multiplicity grows only quadratically. The finite projection and the sorted quadratic inequality are the checked ingredients of Proposition 2.6; grouping the actual infinite tail by height cells and summing the majorant is the argument above.

The bound is used at the endpoint of a window, where the natural index is the positive prime-power jump count strictly below the cutoff. Put

$$j_a = \#\{p^e < 2^a : p \in \{2, 3, 5\}, e \geq 1\} = n_a - 1 \quad (a \geq 1), \quad (11)$$

the equality holding for $a \geq 1$ because the powers of 2 below 2^a number $a - 1$ while the powers of 3 and of 5 below 2^a number $\lfloor \log_3 2^a \rfloor$ and $\lfloor \log_5 2^a \rfloor$; at $a = 0$ the count is $j_0 = 0$ and $n_0 - 1 = -1$. Substituting $n_a = j_a + 1$ into Q gives the integer bound used throughout the rest of the note:

$$K(B, a) = \lfloor B Q(n_a) \rfloor, \quad K(B, a) = \left\lfloor \frac{B(j_a^2 + 10j_a + 27)}{9} \right\rfloor \quad (a \geq 1). \quad (12)$$

The cutoff in (11) is 2^a and it is strict. The symbol $K(B, a)$ is an integer cap for an integral quantity BX_a : from $BX_a \leq BQ(n_a)$ one may take the floor only after integrality has been established. We do not assert $BX_a \leq K(B, a)$ for arbitrary real tails.

Lemma 6.2 (finite denominator clearing). *For all integers $0 \leq u \leq b$ the window mass $h_b \sum_{a=u}^{b-1} s_a$ is a natural number. If $S = N/D$ with $N \in \mathbb{Z}$ and $D \in \mathbb{N}_{>0}$, then $DX_a \in \mathbb{Z}$ for every $a \geq 1$, and there are indices $1 \leq i < j \leq D + 1$ for which $X_i - X_j \in \mathbb{Z}$.*

Proof. An empty window has mass zero. Otherwise $b \geq 1$, and every integer $x < 2^b$ has 2-height exponent at most $b - 1$ while its other height exponents are at most those at 2^b , so $H(x) \mid H_b / 2 = h_b$ and every term of the finite window clears at h_b . Writing $v_a = h_a \sum_{u < a} s_u \in \mathbb{N}$ gives $DX_a = h_a N - Dv_a \in \mathbb{Z}$. Among $D + 1$ of these integers two share a residue modulo D , and the corresponding states differ by an integer. $\square$

The strict upper endpoint is what permits division by two. A cutoff including 2^b would not clear its term at the normaliser $h_b = P_b/2$.

Theorem 6.3 (positive reduced carries from rationality). *Suppose $S = N/D$ with $N \in \mathbb{Z}$, $D \in \mathbb{N}_{>0}$, and write*

$$D = 2^u 3^v 5^w B, \quad u, v, w \in \mathbb{N}, \quad B \in \mathbb{N}_{>0}, \quad \gcd(B, 30) = 1, \quad a_D = u + 1 + 2v + 3w.$$

Then for every $a \geq a_D$ the number $z_a = BX_a$ is a positive integer and

$$z_{a+1} = b_a z_a - B m_a, \quad 1 \leq z_a \leq K(B, a) \leq 90B(a+1)^2.$$

Proof. Let $M = 2^u 3^v 5^w$. For $a \geq a_D$ we have $2^a \geq 2^{u+1}$, $2^a \geq 3^v$ and $2^a \geq 5^w$, using $3 < 2^2$ and $5 < 2^3$; hence $M \mid h_a$. In the clearing identity $DX_a = h_a N - Dv_a$ of Lemma 6.2 both terms on the right are divisible by M , so dividing by M shows that BX_a is an integer. Positivity and the recurrence come from (10), and the upper bound is Theorem 6.1 with the floor taken, since z_a is an integer below $BQ(n_a)$. Finally $n_a \leq 3a$, so $Q(n_a) \leq a^2 + \frac{8}{3}a + 2 \leq 90(a+1)^2$. $\square$

The smooth part M affects the clearing onset; the surviving carry bound depends on B . The displayed a_D is a convenient sufficient onset. The next proposition gives the exact onset when N/D is in lowest terms.

Proposition 6.4 (exact denominators and minimal clearing). *Suppose $S = N/(MB)$ is in lowest terms, with $M = 2^u 3^v 5^w$ and $\gcd(B, 30) = 1$. For every $a \geq 1$,*

$$\text{den}(X_a) = \frac{MB}{\gcd(M, h_a)}, \quad \text{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Hence the first integral reduced tail occurs at

$$a_* = \min\{a \geq 1 : 2^a \geq \max(2^{u+1}, 3^v, 5^w)\} \leq a_D,$$

and BX_a is integral exactly for $a \geq a_$. This onset is computable by integer powers; it is not an estimate obtained by rounding logarithms.*

Proof. Write $X_a = h_a N/(MB) - v_a$, with $v_a \in \mathbb{Z}$ as in Lemma 6.2. Since N is coprime to MB and h_a is supported on $\{2, 3, 5\}$, reduction gives both denominators. Now $M \mid h_a$ means $a-1 \geq u$, $\lfloor \log_3 2^a \rfloor \geq v$ and $\lfloor \log_5 2^a \rfloor \geq w$, precisely the three integer inequalities. All three persist when a increases, giving the first and every later integral reduced tail. The earlier bounds $3 < 4$ and $5 < 8$ give $a_* \leq a_D$. $\square$

This is an ordinary refinement of the supplied sufficient-onset bridge, not a claim of a newly checked Lean theorem.

The integral branch has one further exact property, recorded because it constrains any attempt to build a surviving seed by hand.

Proposition 6.5 (upward closure and rigidity). *For every a , $X_a = (m_a + X_{a+1})/b_a > 0$, and if $X_a \in \mathbb{Z}$ then $X_n \in \mathbb{Z}$ for every $n \geq a$. Moreover, fix A , a positive width function w with $w(A+k)/8^k \rightarrow 0$, and a real orbit $(y_n)_{n \geq A}$ satisfying $y_{n+1} = b_n y_n - m_n$. If y_n and X_n both lie in $(m_n/b_n, m_n/b_n + w(n)]$ for every $n \geq A$, then $y_A = X_A$.*

Proof. The identity is the recurrence solved for X_a , and positivity holds because every shell contains its dyadic left endpoint. Integer coefficients give upward closure. For the last assertion, $y_{A+k} - X_{A+k} = (P_{A+k}/P_A)(y_A - X_A)$. The height quotient satisfies $P_{A+k}/P_A > 8^k/15$ for $k \geq 1$ (Proposition 7.4), whereas the common interval bounds the absolute difference by $w(A+k)$. Thus $|y_A - X_A| < 15w(A+k)/8^k \rightarrow 0$. $\square$

6.1 The jump-constrained improvement

Proposition 6.6 (a smaller bound for the actual tail). *For every $a \geq 0$,*

$$0 < X_a \leq \tilde{Q}(n_a) < Q(n_a), \quad \tilde{Q}(n) = \frac{1210n^2 + 9130n + 18847}{11979}.$$

Proof. For a point $x \geq 2^a$, let d_p be the increase in its p -height exponent from the boundary 2^a , and put $k = d_2 + d_3 + d_5$. Writing $B = \lfloor \log_3 2^a \rfloor$, we have $x < 3^{B+d_3+1} \leq 2^a 4^{d_3+1}$, hence $d_2 \leq 2d_3 + 1$. Thus $j = \lfloor (k+1)/3 \rfloor \leq d_3 + d_5$ and

$$\frac{H(x)}{P_a} \geq 2^{k-j3j} =: d(k).$$

There is at most one height cell of each rank, with at most $(n_a + k + 3)^2/9$ supported points by the earlier projection bound. Consequently

$$X_a \leq \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{18d(k)}.$$

Now $d(3m) = 12^m$, $d(3m+1) = 2 \cdot 12^m$ and $d(3m+2) = 6 \cdot 12^m$. Grouping in threes and summing the quadratic geometric series gives $\tilde{Q}(n_a)$. Finally $11979(Q(n) - \tilde{Q}(n)) = 121n^2 + 1518n + 5111 > 0$ for $n \geq 0$. $\square$

The supplied public Lean at commit 3d6d938d696fed0fb71dd55115a18a73738ff223 proves this for the *actual* normalised tail, not only for an abstract jump model: `actual_sharp_tail_bound` in `ActualSharpTailMajorantR10.lean`, namespace `ErdosProblems.Erdos269.PaperR10` (line 339 in the attached file). The smaller integer cap $\tilde{K}(B, a) = \lfloor B\tilde{Q}(n_a) \rfloor$ is therefore available to the rational direction. We retain K in the main equivalence and old finite certificates to avoid silently changing the recorded tests. No optimality claim is made for either quadratic bound. The rational implication also works for any $G \geq \tilde{K}$; together with $G(B, a) = o(8^a)$, the same proof gives an equivalence on this enlarged band. The named theorem below deliberately retains K so that its formal-source interface and the archived certificates are unchanged.

7 Residue windows and the equivalence band

The bases and accumulated forcing in this section are integer sequences; the actual tails remain real unless rationality has been assumed. For $\ell \geq 1$ and $h \geq 0$, set

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h}W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h}F_{\ell,h} + m_{\ell+h}.$$

Induction gives

$$X_{\ell+h} = W_{\ell,h}X_\ell - F_{\ell,h}, \quad z_{\ell+h} = W_{\ell,h}z_\ell - BF_{\ell,h} \quad (13)$$

for any orbit satisfying $z_{n+1} = b_n z_n - Bm_n$. For integers $C \geq 1, N$, use $\text{lpr}_C(N) = 1 + ((N-1) \bmod C) \in \{1, \dots, C\}$. In particular $\text{lpr}_C(0) = C$. Positivity of this representative is what allows comparison with a positive integral carry.

Proposition 7.1 (the finite residue contradiction). *Let $C > 0$ and let c be an integer with $0 < c$ and $|c| \leq K$. If $c \equiv N \pmod{C}$ and $K < \text{lpr}_C(N)$, then the hypotheses are contradictory.*

Proof. The canonical representative lies in $\{1, \dots, C\}$, and the inequalities put c strictly between 0 and C . If $N \equiv 0 \pmod{C}$ its representative is C while $c \bmod C = c \neq 0$. Otherwise c and $\text{lpr}_C(N)$ are each their own residue and congruence makes them equal, contradicting $|c| \leq K < \text{lpr}_C(N)$. $\square$

For a bound $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$ let $E(G)$ be the statement

$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0 \text{ and } h \geq 1 \text{ with } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > G(B, \ell + h). \end{aligned} \quad (14)$$

The window may depend on both B and a_0 . All quantities in the inequality are finite integers. The quantifiers over denominators and prescribed onsets are nevertheless unbounded; a search over a finite rectangle of (B, ℓ) does not verify them. Every $b_a > 0$, so $W_{\ell,h} > 0$ is automatic.

Theorem 7.2 (the equivalence band). *Let $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$ satisfy $K(B, a) \leq G(B, a)$ for all B and a , and $G(B, a)/8^a \rightarrow 0$ as $a \rightarrow \infty$ for each fixed B . Then*

$$E(G) \iff S \notin \mathbb{Q}.$$

Both K of (12) and $K_0(B, a) = 90B(a+1)^2$ lie in this band, so $E(K)$, $E(K_0)$ and irrationality of S are mutually equivalent. The lower half of the band cannot be dropped: $E(0)$ holds automatically, since every least positive residue is at least 1.

Proof. Suppose $E(G)$ and suppose $S = N/D$ were rational. Theorem 6.3 supplies $B \geq 1$ coprime to 30, an onset a_D , and positive integers $z_a = BX_a \leq K(B, a) \leq G(B, a)$ for $a \geq a_D$ satisfying the cleared recurrence. Apply (14) with $a_0 = a_D$ to obtain a window (ℓ, h) with $\ell \geq a_D$. By (13), $z_{\ell+h} \equiv -BF_{\ell,h}$ modulo $W_{\ell,h}$, while $0 < z_{\ell+h} \leq G(B, \ell + h) < \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h})$. Proposition 7.1 is the contradiction, so S is irrational.

Conversely suppose $S \notin \mathbb{Q}$, and fix $B \geq 1$ coprime to 30 and $a_0 \geq 1$. Set $\ell = a_0$ and $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0, 1)$; the prefix identity makes BX_ℓ irrational. For all sufficiently large h , $0 < \delta + BX_{\ell+h}/W_{\ell,h} < 1$, because $X_{\ell+h} = O((\ell + h + 1)^2)$ and $W_{\ell,h} > 8^h/15$. The integer $\lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h}$ is therefore exactly

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its first term eventually exceeds $G(B, \ell + h)$, since $G(B, a) = o(8^a)$. Thus every sufficiently long window from this fixed start escapes, which is stronger than the required existence.

For the two named bounds, $K \leq K_0$ by Theorem 6.3, both are quadratic in a for fixed B , and $K \leq K$ trivially. Finally $\text{lpr}_C(N) \geq 1 > 0$ always, so $E(0)$ holds whatever S is, while the required domination $K \leq 0$ fails. $\square$

The supplied source records the fixed-cap equivalence and the larger $o(8^a)$ band with domination of K , in separate modules listed in the source concordance. The exact fixed-start residue identity used above is an ordinary refinement of the proof. No fresh formal-system build was performed for this revision.

Proposition 7.3 (the fixed-start residue limit). *For fixed integers $B, \ell \geq 1$, write $R_h = \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h})$. Then*

$$\lim_{h \rightarrow \infty} \frac{R_h}{W_{\ell,h}} = \begin{cases} \lceil BX_\ell \rceil - BX_\ell, & BX_\ell \notin \mathbb{Z}, \\ 0, & BX_\ell \in \mathbb{Z}. \end{cases}$$

In the nonintegral case, $R_h = (\lceil BX_\ell \rceil - BX_\ell)W_{\ell,h} + BX_{\ell+h}$ for all sufficiently large h . In the integral case, $R_h = BX_{\ell+h}$ for all sufficiently large h .

Proof. The preceding argument uses only nonintegrality of BX_ℓ , not irrationality of S , and proves the first identity. If BX_ℓ is integral, the recurrence makes $BX_{\ell+h}$ a positive integer congruent to $-BF_{\ell,h}$. Eventually it is less than $W_{\ell,h}$, so it is the least positive representative. In both cases $BX_{\ell+h}/W_{\ell,h} \rightarrow 0$. $\square$

For a rational value, a start before the clearing onset can therefore admit arbitrarily long escaping windows. This is why the rational contradiction must choose a start after the onset; one early start alone does not rule out rationality.

Theorem 7.2 makes the producer a restatement of Erdős #269 for $P = \{2, 3, 5\}$ throughout the band, so work on it is work on the target. The rational direction needs a bound valid for every actual positive carry. The bound K is one such bound, and a smaller bound proved valid for the actual carries could replace it; the zero bound is invalid, and $E(0)$ holds vacuously. Enlarging the bound to any $G \geq K$ with $G(B, a) = o(8^a)$, in particular to any subexponential bound above K , preserves the equivalence. Equivalence preserves truth and settles nothing about difficulty, so the reformulation may still be the easier representation to attack.

Two exact countermodels rule out further shortcuts. For $(W, F, B) = (6, 4, 1)$ we have $\text{lpr}_6(-4) = 2$, so the canonical residue need not be coprime to the accumulated base. For $(W, F, B) = (60, 47, 37)$ we have $\text{lpr}_{60}(-37 \cdot 47) = 1$, so a fixed window has no denominator-independent lower bound on that residue. The window may therefore depend genuinely on B .

7.1 How fast the window base grows

The crude estimate $W_{\ell,h} \geq 2^h$ proves the smaller $o(2^a)$ band. The exact height formula proves the $o(8^a)$ band above and sets the scale of any finite search.

Proposition 7.4 (window-growth law). *Put $\theta_3 = \log_3 2$ and $\theta_5 = \log_5 2$. For all $\ell \geq 0$ and $h \geq 1$,*

$$W_{\ell,h} = 2^h 3^{\lfloor (\ell+h)\theta_3 \rfloor - \lfloor \ell\theta_3 \rfloor} 5^{\lfloor (\ell+h)\theta_5 \rfloor - \lfloor \ell\theta_5 \rfloor}, \quad \frac{8^h}{15} < W_{\ell,h} < 15 \cdot 8^h.$$

Proof. Telescoping (8) gives $W_{\ell,h} = H_{\ell+h} / H_\ell$, and the displayed formula is that quotient written out. Each floor difference differs from $h\theta_p$ by less than one, and $3^{\theta_3} = 5^{\theta_5} = 2$,

so the 3-factor lies strictly between $2^h/3$ and $3 \cdot 2^h$ and the 5-factor strictly between $2^h/5$ and $5 \cdot 2^h$. Multiplying the three ranges gives the bounds. $\square$

Corollary 7.5 (no bounded-length escape at all starts). *Fix $B \geq 1$ coprime to 30 and $H \geq 1$. Only finitely many starts ℓ admit an escaping window of length at most H against the bound K .*

Proof. A least positive residue never exceeds its modulus, so escape at (ℓ, h) requires $K(B, \ell + h) < W_{\ell, h} < 15 \cdot 8^h \leq 15 \cdot 8^H$. On the other hand $j_a \geq a - 1$, since the powers $2, \dots, 2^{a-1}$ already lie below 2^a , so $K(B, \ell + h) \geq \lfloor B((\ell - 1)^2 + 10(\ell - 1) + 27)/9 \rfloor$, which tends to infinity with ℓ . $\square$

Corollary 7.5 is the exact reason a finite scan cannot approach the cofinal quantifier by widening its denominator range alone. Rearranging its inequality, an escaping window at start ℓ and endpoint $a = \ell + h$ must satisfy

$$3h > \log_2 K(B, a) - \log_2 15,$$

so the necessary search depth grows like $(\log_2 B + 2 \log_2 \ell)/3$. This is a lower bound on the length that can possibly work, and it is not an upper bound on the first length that does. An explicit eventual escape threshold can be obtained from a positive lower bound for $\delta = \lceil BX_\ell \rceil - BX_\ell$ and for $1 - \delta$, together with the displayed tail and window estimates. Such information has not been supplied for the literal series. A necessary lower bound on window length is not a prediction of the distribution of first successful lengths and does not give an upper search bound.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

8 The finite evidence

The three displayed window tuples are directly replayable finite certificates. The larger scan and denominator exclusions below are archived computational reports. Their full execution records are not supplied here; in particular the two large exclusions lack their machine-readable witnesses. They are not theorem inputs, and none supplies the unbounded quantifiers.

8.1 The dyadic-window scan

A *certificate* is a finite tuple of integers recording one instance of the inequality in (14), so a reader can recheck the modular inequality immediately and reproduce the shell

numerators by exact enumeration. The integer-only **dyadic-window checker** constructs the ordered pure-power jumps, the block bases and the block digits from exact multiplicity counts, and is the historical source of the following tuples. The package supplied with this revision independently reconstructs and verifies them, with columns the denominator B , the start ℓ , the length h , the endpoint jump index $j_{\ell+h}$, the window base, the forcing, the residue $R = \text{lpr}_W(-BF)$ and the bound K of (12).

B	ℓ	h	$j_{\ell+h}$	W	F	R	K
1	1	2	4	60	47	13	9
7	1	3	6	360	289	137	95
16	1	4	9	10800	8735	640	352

The first row reads as follows. The window starts at $\ell = 1$ and has length 2, so $W = b_1 b_2 = 6 \cdot 10 = 60$; the accumulated forcing is $F = 47$; and $\text{lpr}_{60}(-47) = 13$, since $-47 + 60 = 13$, which exceeds $K(1, 3) = \lfloor (16 + 40 + 27)/9 \rfloor = 9$. The third row lies outside the domain of (14), since $\gcd(16, 30) = 2$, and is displayed to illustrate the window arithmetic at greater depth.

The archived scan report covers $B \leq 5000$ coprime to 30 and $100 \leq \ell \leq 3000$: 3,869,934 pairs, with reported first escape length at most 18 in a search to length 24. This full scan was not rerun for this revision. Its histogram, retained in Section 10.4, is an archived observation, not a consequence of the window-growth bound. A fresh suite supplies all 2496 witnesses for $1 \leq B \leq 97$, $\gcd(B, 30) = 1$ and $1 \leq \ell \leq 96$, with first escaping lengths from 1 to 10 in a search allowed up to 24. A separate integer-only verifier reconstructs the shells and checks completeness of this rectangle. Neither suite proves escape for unbounded B or for cofinally many starts; Corollary 7.5 rules out covering the latter quantifier with any fixed maximum length.

8.2 Two finite denominator exclusions

Archived window-128 exclusion report (witness unavailable). The archived report asserts the following, which is not used as a proved result in this revision. At window length $L = 128$, launch $a_1 = 10005$ and 64 starts, no rational value of the $\{2, 3, 5\}$ running-LCM series has reduced denominator MB with M a 30-smooth divisor of $2^{10005} 3^{6312} 5^{4308}$, $\gcd(B, 30) = 1$ and $1 < B \leq B_{\max}$, where $B_{\max}$ is the 106-digit integer

$$B_{\max} = 1134599670999687767349520845707093359257353022286558739363600235 \\ 016103207564063373270305324172145281971729,$$

so that $\log_2 B_{\max} = 348.9846 \dots$

The exponent triple $(10005, 6312, 4308)$ is the height exponent triple of $H(2^{10005})$, so the certificate normalises by the full height at its launch. Lemma 6.2 normalises by the half height h_{10005} ; the certificate's smooth family is accordingly the larger one. The recorded quantities are the window product with $\log_2 P = 386.40993 \dots$, the exclusion index $J = 1$ certified from a reduced basis lying inside the per-start budget, an enclosure

of width 9.674×10^{-227} , and a maximum ratio $\max X/W = 0.185997$. The machine-readable certificate and a replay receipt are absent. Consequently this is a provenance record, not a verified denominator exclusion available to the reader. Before restoring a theorem environment, publish the exact enclosure, reduction basis, integer inequalities, engine revision and reproducible command. No Lean declaration is claimed for it.

Archived continued-fraction exclusion report (witness unavailable). The archived report asserts that the normalised tail X_1 has 13,109 certified partial quotients, so it is not rational with denominator at most 2^{22482} , about 10^{6768} .

The stated certification method is a common prefix of the continued fractions of the two endpoints of an interval provably containing X_1 ; the numbers with a given prefix of partial quotients form an interval [19, Lemma 10.1.2, p. 106], so no approximation heuristic enters, and the truncation was independently checked to agree with the direct smooth-number sum as an exact rational. The machine-readable witness and replay receipt are absent; the asserted exclusion is therefore not a theorem input. A replay must supply rational endpoints, a proof that the entire interval contains X_1 , the common continued-fraction cylinder, and a rigorous lower bound for denominators of all rationals in the enclosure. A count of matching partial quotients alone is not that denominator certificate. The recorded statistics are a largest denominator of 22,483 bits, a largest partial quotient of 129,114, a mean partial quotient of 23.4133, observed Gauss–Kuzmin frequencies 0.4208, 0.1665, 0.0917, 0.0575, 0.0391 against the predicted 0.4150, 0.1699, 0.0931, 0.0589, 0.0406, and a Lévy constant of 1.18869 against $\pi^2/(12 \log 2) = 1.18657$. The predicted values are the almost-everywhere frequencies of Gauss’s law and Lévy’s almost-everywhere constant [18, pp. 288–289 and footnote 5]. These statistics describe a finite prefix and do not bear on whether X_1 is a Liouville number, algebraic, or rational with a larger denominator.

The two archived reports concern different finite families: one uses a lattice at a fixed launch and fixed smooth part, the other a continued-fraction enclosure at $a = 1$. Neither has a replayable witness in this packet. Even after verification, each would exclude only its stated denominator range and neither would settle an instance of the problem.

9 The remaining arithmetic questions

9.1 An exact weighted-shift identity for the repeated series

This subsection concerns S , not $\mathcal{D}_{2,3,5}$. Write $P_a = H(2^a)$, $\alpha = S/2$ and $\alpha_t = \sum_{k < t} m_k / P_{k+1}$. The literal coefficient object is Lemma 5.3; its size bound is not a polynomial representation.

Lemma 9.1 (weighted shifts preserve the actual value). *Fix integers $c_0, \dots, c_\sigma$, not all zero, independently of the positive integer shift r . Put*

$$\gamma_{a,t} = \frac{P_t P_{a+1}}{P_{a+t+1}}, \quad D_{r,a} = 15 \sum_{j=0}^{\sigma} c_j \gamma_{a,jr} m_{a+jr}.$$

Then $\gamma_{a,t} \in \{1, 1/3, 1/5, 1/15\}$ and $D_{r,a} \in \mathbb{Z}$. Furthermore, with

$$A_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr}, \quad Z_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr} \alpha_{jr} \in \mathbb{Z},$$

we have the absolutely convergent identity

$$A_r \alpha - Z_r = \sum_{a \geq 0} \frac{D_{r,a}}{P_{a+1}}. \quad (15)$$

One may take $C = 225 \sum_{j=0}^{\sigma} |c_j| \max(1, j)^2$ in $|D_{r,a}| \leq C(a + r + 1)^2$. If J is the largest index with $c_J \neq 0$, then $A_r \neq 0$ whenever $\sum_{j < J} |c_j| 2^{-(J-j)r} < |c_J|$; an empty sum is zero.

Proof. The two-exponents cancel in γ . For $p = 3, 5$, floor addition leaves exponent zero or -1 , proving the four-valued assertion and integrality. For $k < jr$, P_{k+1} divides P_{jr} , so Z_r is an integer. For each t , absolute convergence gives

$$\sum_{a \geq 0} \frac{\gamma_{a,t} m_{a+t}}{P_{a+1}} = P_t \sum_{k \geq t} \frac{m_k}{P_{k+1}} = P_t (\alpha - \alpha_t).$$

The finite linear combination proves the identity. Lemma 5.3, $0 < \gamma \leq 1$ and $a + jr + 1 \leq \max(1, j)(a + r + 1)$ give the stated constant. For $j < J$, $P_{jr}/P_{Jr} \leq 2^{-(J-j)r}$. Dividing A_r by $15P_{Jr}$ and bounding the remaining terms proves the nonvanishing condition, which holds for all sufficiently large r . $\square$

The identity was derived in the earlier editorial structural note and is included here with its complete proof. It is not claimed as an application of an existing fixed-base theorem or as a Lean-checked statement.

9.2 The boundary term in the weighted difference

Let $T_a = \{(j, k) \in \mathbb{N}^2 : w_{j,k} < a + 1\}$ and, for $0 \leq t < 1$, set

$$\omega_a(t) = \prod_{p \in \{3,5\}} p^{\lfloor (a+1)\theta_p \rfloor - \lfloor (a+t)\theta_p \rfloor}.$$

This weight is defined even for pairs outside T_a . For $\nu \geq 0$, put

$$\begin{aligned} \kappa_p(a, t, \nu r) &= \lfloor (a + t + \nu r)\theta_p \rfloor - \lfloor (a + t)\theta_p \rfloor - \lfloor \nu r\theta_p \rfloor, \\ \chi_\nu(a, t) &= 3^{-\kappa_3(a, t, \nu r)} 5^{-\kappa_5(a, t, \nu r)}. \end{aligned}$$

Each κ_p is 0 or 1. We call it a crossing only in this precise floor-addition sense; the moving edge of T_a is a different boundary.

Proposition 9.2 (an exact interior-and-strip decomposition). *For a fixed operator $c_0, \dots, c_\sigma$ and $r \geq 1$, let $E_0 = T_a$ and $E_s = T_{a+sr} \setminus T_{a+(s-1)r}$ for $1 \leq s \leq \sigma$. Then*

$$\frac{D_{r,a}}{15} = \sum_{s=0}^{\sigma} \sum_{(j,k) \in E_s} \omega_a(t_{j,k}) \sum_{\nu=s}^{\sigma} c_\nu \chi_\nu(a, t_{j,k}).$$

For the cubic operator $(1, -3, 3, -1)$, if all these crossing bits vanish, then

$$\frac{D_{r,a}}{15} = -M_0 + 2M_1 - M_2, \quad M_s = \sum_{(j,k) \in T_{a+(s+1)r} T_{a+sr}} \omega_a(t_{j,k}).$$

Thus absence of floor crossings cancels the common interior, but not necessarily the three boundary strips.

Proof. Cancellation of the two upper-boundary exponents gives the pointwise identity

$$\gamma_{a, \nu r} \omega_{a+\nu r}(t) = \omega_a(t) \chi_\nu(a, t).$$

Insert the triangle formula for each $m_{a+\nu r}$ and group by the first triangle containing a pair. A pair in E_s occurs exactly in the terms $\nu \geq s$, proving the first formula. Under the no-crossing hypothesis, $\chi_\nu = 1$. The full cubic sum is zero, while its three successive suffix sums are $-1, 2, -1$, proving the second formula. $\square$

Example 9.3 (cubic cancellation can fail with no floor crossings). Take $a = 0, r = 35$ and $(c_0, c_1, c_2, c_3) = (1, -3, 3, -1)$. Exact integer comparisons give

u	0	35	70	105
$\lfloor \log_3 2^u \rfloor = \lfloor \log_3 2^{u+1} \rfloor$	0	22	44	66
$\lfloor \log_5 2^u \rfloor = \lfloor \log_5 2^{u+1} \rfloor$	0	15	30	45
m_u	1	195	723	1582

The first two rows imply $b_u = 2$ and $\lfloor (u+t)\theta_p \rfloor = \lfloor u\theta_p \rfloor$ for every $0 \leq t < 1, p = 3, 5$. Hence every relevant $\kappa_p(0, t, u)$ vanishes, $\gamma_{0,u} = 1$, and all triangle weights at these four indices are 1. The last row is therefore the count of pairs satisfying $3^j 5^k < 2^{u+1}$, not a floating-point estimate. The strips have sizes 194, 528, 859, so

$$D_{35,0} = 15(-194 + 2 \cdot 528 - 859) = 45 \neq 0.$$

A portable certificate records the count in every j -row and verifies all power inequalities using integers. This single example disproves unconditional cubic cancellation from carry avoidance. It does not exclude a different fixed operator, different shifts, a boundary correction or a sparse-defect statement restricted to a suitable infinite subsequence.

What the Hecke–Mahler comparison does and does not supply. In the published Luca–Ouaknine–Worrell article [21], Definition 5 requires both expanding gaps and polynomial variation of nonzero finite-difference defects. Theorem 6 is the associated fixed-base value criterion; Theorem 8 and Claim 10 establish the hypotheses for polynomial floor sequences. Claim 10 uses exact finite-difference cancellation when no floor crossing occurs. Our m_a is instead a weighted lattice count. Proposition 9.2 separates floor crossings from its moving boundary, and Example 9.3 shows that the latter can survive a cubic difference even with no crossings. The correction preserves the value, but does not prove sparseness.

A possible coefficient target is a fixed operator and shifts $r_n \rightarrow \infty$ for which $\Delta_n = \{a \geq 0 : D_{r_n, a} \neq 0\}$ is infinite and distinct members are at least ηr_n apart, for a fixed $\eta > 0$. Even this would leave the uniform polynomial-variation requirement, for example

$$|D_{r_n, a'}| \leq C((a' - a)^d + |D_{r_n, a}|) \quad (a < a', a, a' \in \Delta_n),$$

with C, d independent of n . The bound $O((a + r + 1)^2)$ is not this relative-variation estimate. There is also a representation issue: the displayed denominator chain is not a fixed-base power sequence. The next proposition shows exactly what is lost in two direct fixed-base recodings. A theorem for the original chain would need a separate proof, including non-cancellation and the height estimates in any Subspace-Theorem argument. The finite set of prime divisors alone does not provide those estimates.

The echoing comparison has a different interface. Kebis, Luca, Ouaknine, Scoones and Worrell [22, Definition 3, Theorem 6 and Claim 7] work with fixed-base series whose coefficients lie in a finite algebraic alphabet, and require non-vanishing weighted mismatch sums. The four-letter radix sequence is not the numerator sequence: the actual m_a is unbounded. A recoding would have to preserve the scalar value, identify its coefficient alphabet and verify those mismatch conditions. No such recoding is proved here.

Proposition 9.4 (what direct fixed-base recoding preserves). *The following identities converge absolutely:*

$$\alpha = \sum_{a \geq 0} \frac{e_a}{30^{a+1}} = \sum_{a \geq 0} \frac{v_a}{8^{a+1}}, \quad e_a = \frac{m_a 30^{a+1}}{P_{a+1}}, \quad v_a = \frac{m_a 8^{a+1}}{P_{a+1}}.$$

Here $e_a \in \mathbb{Z}_{>0}$ and $e_a \geq (15/4)^{a+1}$, whereas $v_a \in \mathbb{Z}[1/15]$ and $0 < v_a < 225(a+1)^2$. For an integer $q \geq 2$, the termwise divisibility $P_n \mid q^n$ for every $n \geq 1$ holds exactly when $30 \mid q$; that direct recoding then has coefficients at least $(q/8)^{a+1}$.

Proof. Each exponent in P_n is at most n , so $P_n \mid 30^n$. Also $8^n/15 < P_n \leq 8^n$ by the exact two-exponent and the strict three- and five-power bounds. These facts, $m_a \geq 1$ and $m_a \leq 15(a+1)^2$, give all the coefficient bounds and identities. Necessity of $30 \mid q$ follows because each of 2, 3, 5 divides some P_n ; sufficiency follows from $P_n \mid 30^n$. The general lower bound follows again from $P_n \leq 8^n$. $\square$

The base-30 coefficients are integral but not polynomially bounded, so they do not satisfy the hypotheses of the cited fixed-base polynomial-growth criterion. The base-8 coefficients have polynomial size, but integrality and the required arithmetic height bounds are not established. No claim is made that their reduced denominators are unbounded: cancellation with m_a must be considered. These are obstructions to the direct recodings, not a proof that all possible fixed-base representations fail. Adamczewski–Bugeaud’s complexity theorem [25, Theorem 1] concerns the actual digits of an integer-base expansion: for an algebraic irrational their length- n block complexity divided by n tends to infinity. Neither (b_a) nor the uncarried (e_a) is such a digit expansion of α .

Problem 9.5 (the repeated three-prime target). Prove irrationality of $S = \mathcal{R}_{\{2,3,5\}}$. Equivalent forms are $E(K)$ and

$$BX_a \notin \mathbb{Z} \quad \text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a \geq 1. \quad (16)$$

The first formulation uses finite integer data but unbounded denominator and onset quantifiers; the second uses the actual real tail. Neither is an additional hypothesis imposed on the series, and neither is proved by the rank obstruction. The next proposition records their common scalar target, complementing Theorem 7.2.

Proposition 9.6 (the reduced-tail question is also the target). *Statement (16), quantified over every $B \geq 1$ coprime to 30 and every $a \geq 1$, is equivalent to irrationality of S .*

Proof. If $BX_a \in \mathbb{Z}$ for some such B and a , then $X_a \in \mathbb{Q}$, and since $S = \sum_{j < a} s_j + X_a/h_a$ with h_a a positive rational and the prefix a finite sum of rationals, $S \in \mathbb{Q}$. Conversely if $S \in \mathbb{Q}$, then Theorem 6.3 produces B coprime to 30 with $BX_a \in \mathbb{Z}$ for every $a \geq a_D$. $\square$

Integrality at one index propagates forwards, but nonintegrality at one early index does not propagate forever: a rational denominator can clear later. Proposition 6.4 describes that onset exactly. The scaled dichotomy also permits an integral branch, so it does not settle the target.

9.3 The analytic route and what it would need

For the de-duplicated series put

$$\mathcal{D}_{2,3,5} = 1 + \sum_{t \in \{2^n, 3^n, 5^n : n \geq 1\}} \frac{1}{2^{\lfloor \log_2 t \rfloor} 3^{\lfloor \log_3 t \rfloor} 5^{\lfloor \log_5 t \rfloor}},$$

whose dyadic coding is the joint rotation word $\delta_{3,a} = \lfloor (a+1)\theta_3 \rfloor - \lfloor a\theta_3 \rfloor$ and $\delta_{5,a} = \lfloor (a+1)\theta_5 \rfloor - \lfloor a\theta_5 \rfloor$, with $b_a = 2 \cdot 3^{\delta_{3,a}} 5^{\delta_{5,a}}$.

Problem 9.7 (function-faithful two-dimensional representation). The historical irrationality assertion for $\mathcal{D}_{2,3,5}$ is not being reclassified as a new open problem. A recovered proof or a transcendence theorem would require its own argument. For a functional approach, first specify the function, its coefficient field and convergence domain, then prove an exact identity for $\mathcal{D}_{2,3,5}$, and only then apply a stated value theorem. A conditional theorem must display the extra nondegeneracy assumption; a no-representation theorem must specify the class it excludes.

Individual irrationality of θ_3 and θ_5 is what Theorem 4.3 uses, and it is weaker than the joint equidistribution or the rational independence of $1, \theta_3, \theta_5$ that a two-dimensional value theorem may need; the stronger hypothesis is not assumed anywhere above. The finite-observer formalisation isolates the precise faithfulness requirement: equality in a finite observer must imply equality after symbolic realisation, and a genuine finite-dimensional factorisation forces the realised symbolic span to be finite-dimensional ([residue-coboundary form](#), [symbolic realisation](#), [checked](#)), with the carry residue and residue digit confined to their declared intervals ([carry interval](#), [digit interval](#)). No theorem here proves that the literal realised span is infinite.

Which value-theorem hypotheses remain missing. Pellarin’s rank-one theorem [26, Theorem 1.1] treats Hecke–Mahler values with a quadratic irrational slope and a specified rank-one module condition on the algebraic evaluation points. Two lattice coordinates alone imply neither hypothesis. For linear Mahler systems in one variable, Adamczewski and Faverjon give a new proof of Nishioka’s theorem [27, Theorems 1 and 2]; one must have a Mahler functional system and a regular algebraic evaluation point. Their multivariate theory is now published [28]. The author manuscript’s Theorem 3.3 and Sections 3 and 5 retain a genuine functional system, regularity of the evaluation point and admissibility of the transformation–point pair. No such system or evaluation data for the literal three-prime sum are constructed in this record. In particular, the theorem’s freedom in the system matrix does not remove its transformation and point hypotheses. The source audit distinguishes the 68-page author manuscript used for these locators, the older 52-page arXiv version and the journal metadata.

9.4 Structural constraints already in place

The radix alphabet extends without difficulty. For ordered primes $p_1 < \dots < p_s$ an interval (p_1^a, p_1^{a+1}) contains at most one power from each other channel, because consecutive p_i -powers have ratio $p_i > p_1$, so its block radix belongs to the 2^{s-1} -letter alphabet $\{p_1 \prod_{i=2}^s p_i^{\varepsilon_i} : \varepsilon_i \in \{0,1\}\}$. The quantitative questions are the interesting ones: effective recurrence or discrepancy for the actual four-letter $\{2,6,10,30\}$ word, an asymptotic with an error term for the restricted two-dimensional shell counts that generate m_a , or the exact separated rank of the literal kernel under a specified family of shifts.

Three-channel rigidity and carry-lift extinction already exclude one proposed argument in four exact steps. Under channel surjectivity, ordinary block nullity is equivalent to the perturbation being a coboundary of a channel potential (potential classifier). Zero perturbations on genuine $2 \rightarrow 3$ and $2 \rightarrow 5$ transitions then identify all three potential values and force the perturbation to vanish at every index. For an integral lift, that vanishing makes a nonzero initial error grow by the exact product of the successive bases, and bases at least two make its absolute value at least 2^N , contradicting even a single index- N bound strictly below 2^N (one-index extinction); consequently no uniform bound on the lift error can hold either (uniform extinction). A separate four-state calculation reaches the obstruction earlier: four real states in $(0,1)$ with unit-accuracy integral lifts and the two anchor equalities force the first complete 2-block sum to be 1, so that block cannot be null (first-block sum, four-state obstruction).

These lift conclusions are conditional. The paper constructs the actual orbit and its bounded integral carry under rationality, and it does not construct a faithful lift with the two anchors or the block nullity those auxiliary statements require. The actual carry supplies a weighted block defect instead, so any successful argument along that line must use the weighted identity or construct a different faithful lift.

A single conditional single-channel criterion is also on record. For the pure 2-channel sum the Cantor states lie in an explicit open interval (confinement), consecutive states are separated (gap bound), a small nonzero linear form forces irrationality (criterion), and clearing together with small gaps assembles to irrationality (assembly).

The clearing and small-gap hypotheses are not discharged for any concrete series, so no irrationality statement follows from them here.

9.5 Where the problem stands

The two-prime transcendence deduction is an earlier application of the cited value theorem. For the repeated $\{2, 3, 5\}$ value, this record establishes the literal coefficients, integral recurrence, exact clearing, quadratic bounds and the equivalent residue target. It does not establish irrationality. The rank and uniform-norm obstructions concern the kernel, not the arithmetic of its sum. The fixed-start residue formula clarifies the target's quantifiers, while the strip decomposition identifies a concrete obstruction to transplanting cubic no-crossing cancellation. A further argument must use the actual weighted multiplicities to exclude eventual integral reduced tails. Finite tests, countability, low-complexity radix coding and denominator support do not by themselves do so.

Statements and declarations

Proof sources. The two-prime deduction rests on the cited external value theorem and is not formalised here; Fan's recorded priority is retained. The attached `LEAN_INDEX.json` at public commit `3d6d938d696fed0fb71dd55115a18a73738ff223` records a successful CI build for the relevant public source closure. This includes arbitrary-order kernel rank, the actual tail recurrence, rationality clearing, the actual Q and $\tilde{Q}$ bounds, and the $o(8^a)$ equivalence band. The separate Palomar release at `52f29ad1` selects the arbitrary-order uniform-minor and prime non-separation clauses, as well as the finite fixture. Its Comparator evidence is therefore not limited to a 2×2 determinant. These statements report the supplied receipts; this editorial revision did not rerun Lean, Isabelle, Comparator or NanoDa. The source index is not a blanket axiom audit. Individual historical links keep their own pins and must not be treated as byte-identical to the newer tree without a comparison. The weighted-triangle and weighted-shift identities, exact denominator formula, scaled dichotomy, strengthened rigidity, residue limit, strip decomposition and direct recoding proposition have ordinary proofs. The new finite checks support their finite instances, not their infinite quantifiers. None is presented as a new formalisation result of this pass. The large numerical exclusions remain archived reports until their witnesses are published. No evidence item supplies the source-specific cofinal escape.

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, the library manifest, and the exact dyadic-window checker used in the finite scan. Those sources support only the formal statements identified in the evidence register. Ordinary proofs and external analytic inputs have their own stated dependencies; the unavailable numerical witnesses are not supplied by a repository link.

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Acknowledgements. Priority for the two-prime factorisation, its Hecke–Mahler reduction and the running-LCM identity for every $|P|$ belongs to Steve Fan's forum post of 26

June 2026 [12]. The transcendence input is due to Yann Bugeaud and Michel Laurent and to the earlier work of Loxton and van der Poorten cited by them. The problem numbering and historical status snapshot are taken from the Erdős Problems catalogue maintained by Thomas Bloom [4].

10 Extended record: migrated arguments and finite evidence

This part of the record holds material that the short note compressed or moved. It separates auxiliary arguments, historical reports and exact source coordinates. The mathematical references are this record's labels; archived computations are not promoted to theorem inputs by appearing in an appendix.

10.1 The historical and catalogue record

In the primary 1988 source Erdős states the infinite- P assertion as a simple exercise and presents persistence for a finite number of primes greater than one as a probable extension, and not as a theorem [2, p. 106]. The catalogue snapshot cited in the supplied manuscript records an open problem [4]; that historical status is not inferred from the conjectural wording and has not been reverified against the live page for this revision. The 1974 letter writes “given primes $p_1, \dots, p_r$,” without explicitly restricting r , so the singleton case, whose value is $p/(p-1)$, must be excluded by hand; the modern restriction $|P| \geq 2$ does that.

The letter's de-duplicated assertion is made for a general finite list of primes and supplies no proof, so it is an asserted historical result and not an open problem. The note therefore restricts its open statement to the repeated series $\mathcal{R}_p$. A recovered proof of the letter's assertion, or a transcendence statement for $\mathcal{D}_p$ with $|P| \geq 3$, would be a different question and needs its own formulation.

The letter prints no argument. On 26 June 2026 Steve Fan posted the two-prime factorisation, the Hecke–Mahler reduction and the transcendence conclusion in the discussion thread of the problem's page [12]; the comment itself notes that the argument does not seem to generalise immediately to $|P| \geq 3$, and a reply there observes that it applies to arbitrary coprime pairs. The supplied publication record dates the note's first public manuscript to 22 July 2026, at commit a9d3ab8, after Fan's post. We retain the calculation as exposition and make no priority claim for it.

The statement of the problem has been formalised as a conjecture with an unfilled proof in the *Formal Conjectures* collection [5]. Its Nat-indexed series includes the empty-prefix least-common-multiple term, so its value differs from the conventional one by a rational constant; transporting a theorem across that boundary needs an explicit series-identification lemma, which is not supplied here.

10.2 The abstract carry lemmas in their general form

The note applies the carry machinery to the literal $\{2, 3, 5\}$ orbit. The underlying statements are about integer sequences alone, they are reusable, and they are recorded here in the generality in which they are checked. They are conditional carry lemmas in the Cantor-series tradition described after Theorem 5.2, and each assumes its divisibility or escape hypothesis.

Let $D = D_{\text{sm}}B$ with $D_{\text{sm}} = 2^u 3^v 5^w$ and $\gcd(B, 30) = 1$, and let (c_n) be an integer sequence satisfying $c_{n+1} = b_n c_n - D m_n$ for an integer radix word (b_n) and forcing word (m_n) .

Proposition 10.1 (conditional denominator reduction). *If $c_n = D_{\text{sm}} d_n$ for every n , with $D_{\text{sm}} > 0$, then the recurrence, positivity, upper bound and window identity for (c_n) reduce to the same four statements for (d_n) with multiplier B in place of D .*

Height absorption alone does not imply divisibility of a carry state. The identity $DX_a = h_a N - Dv_a$ of Lemma 6.2 is what supplies it for the actual orbit, and Theorem 6.3 is the instance in which the note uses it. The formal consumer takes the common-factor form as a hypothesis.

Proposition 10.2 (conditional extinction of bounded carries). *Let (b_n) and (m_n) be a radix word and a forcing word, let $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$, and assume cofinal local-window escape against G . Fix $B > 0$ coprime to 30. There is no integral sequence (d_n) satisfying simultaneously $d_{n+1} = b_n d_n - B m_n$, $d_n > 0$ and $|d_n| \leq G(B, n)$ for every $n \geq 0$.*

Proof. Choose one escaping window (ℓ, h) . The window identity gives $d_{\ell+h} \equiv -BF_{\ell,h}$ modulo $|W_{\ell,h}|$. The endpoint state is positive and at most $G(B, \ell + h)$, whereas the canonical positive residue of the right-hand side exceeds that bound, so Proposition 7.1 applies. $\square$

Coprimality with 30 is used by the escape hypothesis to select a window; once a window is fixed, the finite contradiction does not use it. The formalisation carries the edge cases: a zero window base is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint carry from vanishing. The packaged absorbed form takes a nonzero smooth factor, the exact factorisation $c_n = D_{\text{sm}} d_n$, the absorbed recurrence for (c_n) and the positive short bound for (d_n) , and derives the same contradiction in one statement.

10.3 Worked finite examples

The ten smallest values of the running least common multiple at $\{2, 3, 5\}$ are tabulated in Section 2. They illustrate both parts of Proposition 2.2: the value is constant on $\{5, 6, 7\}$ and on $\{9, 10\}$, and each change multiplies by a single prime, by 2 at $x = 2, 4, 8$, by 3 at $x = 3, 9$ and by 5 at $x = 5$. Also $L(10) = 8 \cdot 9 \cdot 5 = 360$ is the least common multiple of the smooth numbers 1, 2, 3, 4, 5, 6, 8, 9, 10.

For Proposition 2.4 take $(p, q, r) = (2, 3, 5)$ and the box $\mathcal{B}(1, 1, 1)$, whose eight points carry the smooth values 1, 2, 3, 5, 6, 10, 15, 30 and the heights

$p^i q^j r^k$	1	2	3	5	6	10	15	30
H	1	2	6	60	60	360	360	10800

Six heights occur, two of them twice: the points 5 and 6 share the height 60, and 10 and 15 share the height 360. The identity reads

$$1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{60} + \frac{1}{60} + \frac{1}{360} + \frac{1}{360} + \frac{1}{10800} = 1 + \frac{1}{2} + \frac{1}{6} + \frac{2}{60} + \frac{2}{360} + \frac{1}{10800} = \frac{18421}{10800},$$

and the two coefficients 2 carry the whole content of the regrouping on this box. Where the heights are pairwise distinct the identity is a relabelling.

Multiplying block radices along a run of blocks gives the product of the jump-word letters over that run: for instance $b_1 b_2 = 60$ is the product of the four multipliers at 3, 4, 5, 8. Block 4 is the only one among the first six carrying an internal power in both channels, and block 5 carries neither, so its radix falls back to the terminal factor alone.

10.4 The finite-scan histogram

The archived scan report described in Section 8 claims coverage of $B \leq 5000$ coprime to 30 and $100 \leq \ell \leq 3000$, with search depth 24. It reports 3,869,934 pairs, an escape in every case, and the following first-success histogram. These figures were not rerun for this revision and are retained as historical data:

h	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
#	1	104	812	5437	51409	237423	735450	1431226	1132756	236752	34910	3076	521	49	8

The report's first case at its maximal observed length 18 is $B = 917$ at start $\ell = 2980$, with endpoint jump index 6179, window base 18139852800000000, forcing 13196471407660025821045, residue 76322101735 and bound 3896420420. An earlier archived run of the same checker over $B \leq 1000$ coprime to 30 and $100 \leq \ell \leq 500$ reports 106,666 pairs with maximal first successful length 14, whose first case is $B = 359$ at start 291, endpoint jump index 627, base 5038848000000, forcing 25864575212865807, residue 213175287 and bound 15932659.

These histograms concern a bounded region. The window-growth law gives a necessary lower bound on a possible escape length, not a distribution law for residues or first successful lengths. No generic-residue model is used as evidence for cofinal escape.

10.5 Source inventory

Each entry names an informal statement and the Lean declaration that carries it. Each link is pinned to its own immutable revision; the exact-reference inventory records the revisions and their comparison with the current public source commit.

informal statement	linked Lean source
smooth lattice value	smooth3 val
pure-power height	three prime height
lattice kernel	three prime kernel q
smooth prefix index set	smooth prefix exponents
running least common multiple	smooth prefix lcm
prefix value divides the height	smooth3 val dvd three prime height of mem
divisibility into the height	smooth prefix lcm dvd three prime height
first pure-power membership	pure first mem smooth prefix exponents

informal statement	linked Lean source
second pure-power membership	pure second mem smooth prefix exponents
third pure-power membership	pure third mem smooth prefix exponents
running-lcm identity	smooth prefix lcm eq three prime height
cell relation	same three prime log cell
cell constancy of the height	three prime height eq of same log cell
cell constancy of the running value	smooth prefix lcm eq of same log cell
cell constancy of the kernel	three prime kernel q eq of same log cell
positive power sets	positive prime powers
channel cardinality	positive prime powers card
exclusion of the origin	one not mem positive prime powers
channel disjointness	positive prime powers disjoint
positive jump channels	three prime positive jump set
positive jump count	three prime positive jump set card
jump set with the origin	three prime jump set with origin
jump count with the origin	three prime jump set with origin card
first height step	three prime height first log step
second height step	three prime height second log step
third height step	three prime height third log step
first coordinate step	smooth prefix lcm first log step
second coordinate step	smooth prefix lcm second log step
third coordinate step	smooth prefix lcm third log step
exponent box	smooth exponent box
point height	smooth point height
height fibre	smooth height fiber
fibre sum	smooth height fiber kernel sum
height-fibre normal form	finite smooth kernel sum grouped by height
cubic majorant	three prime height le cube
origin value	kernel 235 origin
value at two	kernel 235 two
value at three	kernel 235 three
value at six	kernel 235 six

informal statement	linked Lean source
non-separation witness	kernel 235 not rank one
variable-base tail step	tail state step
expanded tail step	tail state step eq
smooth exponent shell	smooth exponent shell
short-interval uniqueness	exponent unique in short interval
first-coordinate projection	smooth exponent shell card le drop first
third-coordinate projection	smooth exponent shell card le drop third
sorted quadratic estimate	sorted pair quadratic
quadratic shell bound	smooth exponent shell card quadratic
dyadic internal power	dyadic internal power
internal-power uniqueness	dyadic internal power exponent unique
dyadic block base	dyadic block base235
exact dyadic radix alphabet	dyadic block base235 cases
bounded-radix consequence	dyadic block base235 mem interval
rank-two certificate	kernel 235 minor eq neg one fifteen
no integer rotation orbit	no integer orbit logb of prime
height factorisation	three prime height factorisation
kernel factorisation	three prime kernel q factorisation
two-prime outer product	two prime kernel q eq outer product
two-prime vanishing minors	two prime kernel q minor two eq zero
uniform minors, general form	exists uniform nonsingular three prime kernel minor
uniform minors for primes	exists uniform nonsingular three prime kernel minor of prime
no finite separation	not finite separable three prime kernel
second-order minor	kernel 235 minor2 eq neg one fifteen
third-order minor	kernel 235 minor3 eq one over 81000
misleading proportional row	kernel 235 row three eq smul row zero
failure of that proportionality	kernel 235 row three ne smul row zero at four
ordered block digit	dyadic ordered block digit235
normalised state step	dyadic normalized tail state r235 succ
shell summability	summable dyadic shell mass r235
infinite tail recurrence	dyadic normalized shell tsum tail r235 succ

informal statement	linked Lean source
integer-or-cofinally-far chotomy	di- dyadic shell tsum tail integer or cofinal far
boundary divisibility	smooth height mul prime dvd boundary height
half-height identity	two mul height normalizer235
window clearing identity	height normalizer235 mul window mass eq int
all-scale rationality lattice	qsmul normalized tail state eq int of value eq rat
normalised-state collision	exists normalized tail state collision of value eq rat
least positive residue	least positive residue
positive representative range	least positive residue pos le
representative congruence	least positive residue mod eq
escape condition	residue escapes window
finite natural-state contradiction	no bounded positive state of residue escape
contrapositive form	residue le bound of bounded positive state
integer least-positive-residue obstruction	no bounded positive int state of least positive residue
exact residue classifier	least positive residue eq nat abs of pos le mod eq
window base	window base
window forcing	window forcing
affine window identity	affine recurrence window
scaled forcing identity	window forcing const mul
integral carry window	integral carry window
endpoint residue identifica- tion	least positive residue window forcing eq carry
absorbed smooth factor	smooth3 val dvd three prime height of le
common-factor cancellation	integral carry cancel common factor
reduced bound transfer	reduced carry pos le of common factor
reduced window transfer	reduced integral carry window
cofinal window hypothesis	cofinal local window escape
reduced-carry extinction	no positive reduced carry of cofinal local window escape
absorbed-carry extinction	no positive absorbed carry of cofinal local window escape
real window identity	true normalized state window

informal statement	linked Lean source
escape from irrationality, general bound	cofinal local window escape of irrational of beaten
escape from irrationality, quadratic family	cofinal local window escape of irrational of quadratic
escape from irrationality, actual bound	cofinal local window escape of irrational
producer equivalence, shifted tail	actual cofinal local window escape iff
producer equivalence, series value	actual cofinal local window escape iff irrational value
rationality-to-carry bridge	exists reduced carry of value eq rat
the actual producer	actual cofinal local window escape
irrationality from the producer	irrational value of cofinal local window escape
bounded-radix alternative	bounded radix zero or cofinal far
rational value from an integral scaled tail	rational of scaled tail integer
channel potential classifier	channel block null iff channel potential
one-index lift extinction	no carry lift of error bound below two pow
uniform lift extinction	no bounded carry lift of block null two anchors
first-block sum	binary lift two anchors force first block sum one
four-state obstruction	no unit accurate lift with two anchors and first two block null
residue-coboundary form	carry eq residue digit add coboundary
carry interval	carry residue mem interval
digit interval	residue digit mem interval
symbolic realisation	cartier realisation
finite realised span	finite realised span of factorisation
Cantor-state confinement	state mem ioo
state gap bound	state gap lower bound
small-form criterion	irrational of small nonzero forms
conditional assembly	irrational of clearing and small gaps
staircase index engine	staircase indices

11 Mathematical result and evidence map

The old corpus-wide ranking and result-population metadata are preserved verbatim in the accompanying archive, not treated as mathematical evidence. The reader-facing dependencies are as follows.

Result	Dependency and limit
Height and cells	Prime-power divisibility and unique factorisation. Their geometry alone gives no irrationality statement.
Two-prime values	Earlier factorisation plus the cited external Hecke–Mahler value theorem; not formalised here.
Arbitrary-order rank	Diagonal rescaling, separate one-dimensional densities and a staircase determinant; selected supplied formal evidence.
Actual tail	Literal multiplicities, half-height normalisation and convergence; an abstract recurrence would not identify the value.
Rationality bridge	Strict endpoint clearing and a valid integral carry cap. Exact onset is an ordinary refinement.
Window equivalence	The actual recurrence, valid cap and octic window growth; the cofinal escape itself is unproved.
Boundary analysis	Exact weighted reindexing and strip decomposition; the finite cubic counterexample has a replayable integer witness.
Finite evidence	Individual checked witnesses certify only their stated windows; large missing-witness reports remain archival.

A source location, a successful build receipt, a selected formal statement, an axiom audit and a finite execution receipt answer different questions. No global registry population or editorial rank supplies any of them.

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